Bypass Flow Via Root-Induced Macropores (RIMS) in Subirrigated Agriculture

A.J.M. Smucker, W. Richner, and V.O. Snow, Crop and Soil Sciences Department, Michigan State University, East Lansing, MI 48824. Phone: (517) 355-8370. FAX: (517) 353-5174.

Summary

Increasing competition for a limited supply of freshwater requires that groundwater supplies be maintained free of nitrates, pesticides, and other toxic chemicals related to agriculture. Greater application rates of nitrogen fertilizers and pesticides to intensive agricultural production systems have caused increased quantities of nitrates and other chemicals to appear in the drinking waters of many communities proximal to agricultural farmlands. Many of these agricultural pollutants move through the root zone of agricultural crops by preferential pathways, bypassing the normal transport of soil solutions through smaller and more slowly conducting soil pores. Large soil pores consisting of soil cracks, earthworm channels, and root-induced macropores often directly connect the soil surface with regions deep within the soil. Managing these larger pathways would greatly reduce the contamination of most groundwater supplies.

Plant root systems of row crops absorb large quantities of soil nitrogen during their exponential growth stages. However, certain forms of nitrogen are lost to the soil through the decomposition of plant roots and other plant residues. Not all soil nitrates are absorbed during the growing season. In fact, those crops which are planted on medium- and fine-textured soils, e.g., potatoes, absorb as little as one-half of the nitrogen fertilizers applied at planting. Nitrates from the unused fertilizer and from the mineralization of soil organic matter become very mobile within the soil profile, especially during the periods of time when soil water contents are high. In the late fall, winter, and early spring, when soil water contents approach saturated conditions, there is considerable movement of nitrates within soil profiles to depths beyond the root zones. Planting of cover or "catch" crops, whose living root systems absorb additional nitrates and plug some of the many large soil macropores, can bridge the time between cash crops when nutrient leaching potentials are the greatest. Additionally, newly engineered plant species with expansive root systems which require greater quantities of photo-assimilated carbon are being developed to intercept greater quantities of nitrates and other potentially harmful metals within the soil solution.

Preferential flow is an extremely important factor in the leaching of solutes through heterogeneous soils. However, it is not often realized that preferential flow can either increase or decrease solute leaching (Helling and Gish, 1991). A decrease in solute leaching would occur, for example, when the solute is sequestered in the immobile soil water associated with soil aggregates. Thus the microdistribution of the solute between the mobile and immobile soil water is of great importance. Current research programs are investigating new methods which improve the adsorption and biodegradation of these soil pollutants.

Subirrigation of plants through the tile lines, installed to drain wet soils in the springtime, provides an economic opportunity for supplying water to the root systems of agricultural plants without large capital investments for new irrigation equipment. Water table management systems, which combine subirrigation with subsurface drainage, provide opportunities for increasing profit. Additionally, subirrigation provides an excellent new approach for optimizing concentrations of soil nitrogen within the root zone, reducing the delivery of nitrates from croplands to receiving waters beyond the reach of plant root systems.

Methods

Laboratory, greenhouse, and field studies[1] have been conducted at Michigan State University which address the question of whether living plant root systems will intercept some of the mobile nutrients in soil solutions and reduce the movement of these nutrients in saturated soils by physically plugging some of the larger soil pores. The topsoil, Ap, horizon of a Kalamazoo loam (fine loamy mixed mesic Typic Hapludolf) soil was sieved to a uniform range of medium soil aggregates, removing organic plant debris and natural macropores from the soil column. Each soil column was packed to a uniform bulk density in large columns (7.8-inch diameter x 15.7-inch height). The soil in each column was saturated and the saturated hydraulic conductivity (K_S) measured and drained. Canola, corn, and ryegrass were planted in 8 replications. One treatment contained no plants. All plants were harvested after 8 weeks of plant growth and the K_S values were measured for each column. Soil water contents were maintained at near field capacity to promote root decomposition for 8 additional weeks and K_S measurements were determined again and the columns drained. Then, one-half of the columns were subjected to a methylene blue dye for several days. The stained columns were separated into 4-inch sections for observation and further measurements were made on each soil depth. Columns with previous plants and no dye were replanted to a ryegrass cover crop for a 4-week period. Ryegrass was harvested and Ks measurements were taken and the soil drained. These columns were also stained and separated into 4-inch sections for observation and additional measurements.

Results

Live roots of canola and corn increased the saturated hydraulic conductivity (K_S) of the Kalamazoo loam soil by 50% and 100%, respectively (Fig. 1). Ryegrass roots reduced K_S by 30% during the same 8-week period of plant growth. Death and decomposition of corn and ryegrass roots, during the 8-week period following an early harvest, increased K_S values by 380% and 220% of the unplanted soil. Decomposing canola roots appeared to have little influence on K_S values as water movement in the saturated soils changed very little when planted to canola (Fig. 1). Cover crop of ryegrass, planted after 8 weeks of root decomposition, greatly reduced the K_S values of this loam soil to rates nearly equal to the unplanted soils, 4 weeks after planting (Fig. 1). Roots of ryegrass, following corn, reduced Ks values to 100% of the unplanted soil. Conductivities of the unplanted soil remained nearly constant for the duration of the 20-week study. Root plugging of greenhouse soils are currently being tested in field studies during the 1994-95 growing seasons.

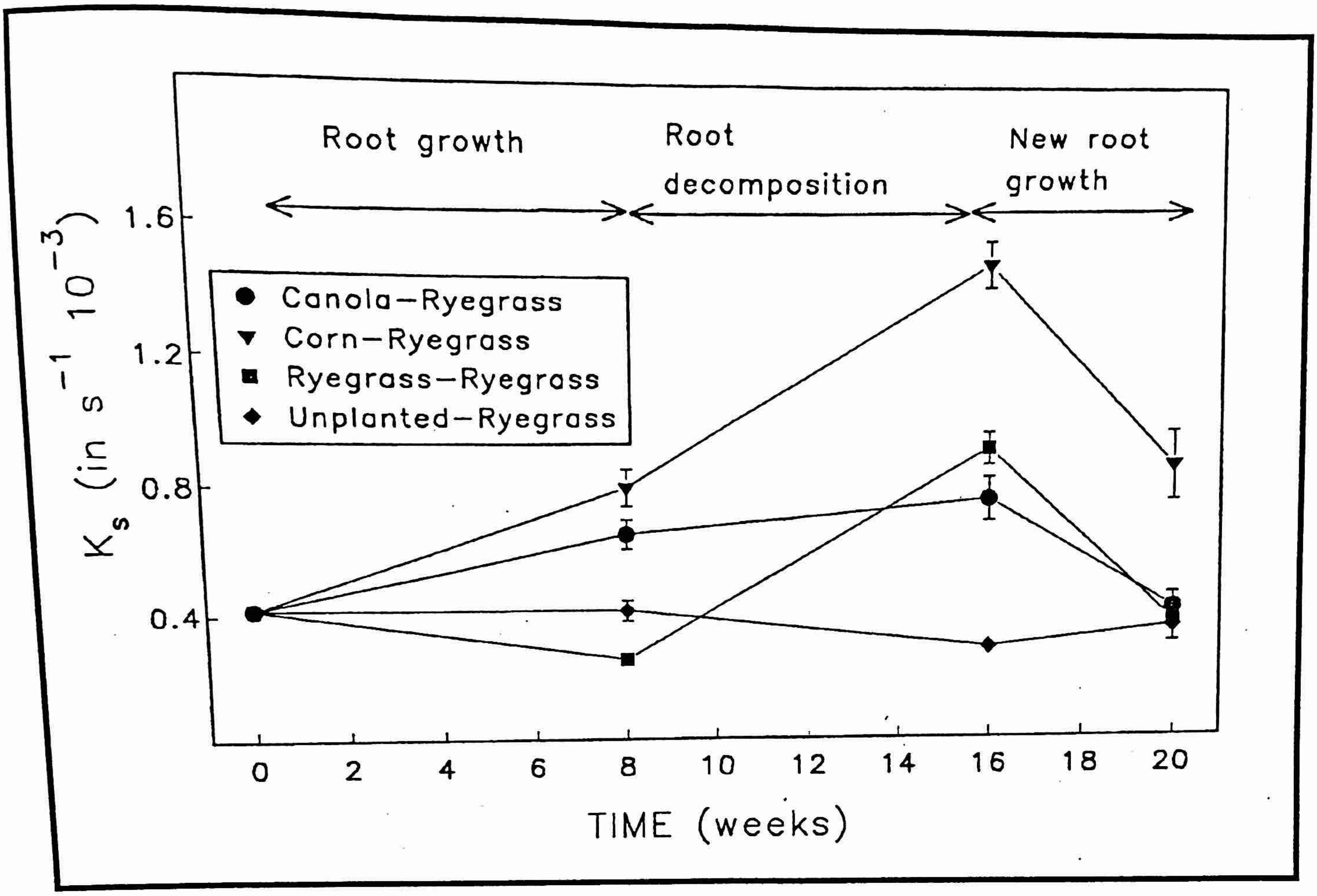

Figure 1. Saturated hydraulic conductivity (K_S) responses to root growth, death and regrowth of three crop plants in a Kalamazoo loam soil. (Vertical bars represent S.E.)

The migration of dyes through these soil columns suggested that roots of corn facilitate the establishment of a more continuous pathway for water flow via root-induced macropores (RIMS) than do canola and ryegrass crops, resulting in greater saturated water flow through the RIMS produced by corn. Bromide pulse-labeling of the saturated soils and subsequent breakthrough curves indicated greater transport of ions through soils containing RIMS. Convection-dispersion equations are being used to quantify the exchange of ions from the mobile to immobile phases of the solutions in soils with and without RIMS which have been developed by different crop species and root regrowth patterns.

Subsequent field measurements of the K_S values for natural soils suggest that up to ten-fold greater quantities of water move through saturated soils than through the repacked greenhouse columns, reported above. Higher K_S rates in natural soil profiles result from the greater numbers of macropores which are formed by earthworms, plant roots, and other soil organisms. These data suggest the need for incorporating cover or catch crops into fields following most cash crops.

Discussion

The greater numbers of freshwater wells containing higher concentrations of nitrates among a growing number of agricultural communities require modifications in the application rates of fertilizers, improved crop utilization, and a reduction of nitrate leaching within sustainable agricultural production systems. Leaching of nitrates, some pesticides, and other soluble soil toxins can be reduced by minimizing the flow of water through the soil profile. Most water flow through the soil profile occurs when the soil is nearly saturated. Soils become saturated more frequently when crops are absent as plants absorb and transpire large quantities of soil water. The incorporation of additional crops within current crop rotations, especially during seasons when plant growth is reduced by colder climates, appears to be a strategic approach for reducing the flow of nutrients into deep groundwater supplies. Data reported here suggest that a corn crop greatly increases or actually decreases the rate of water movement through wet soils. Greater quantities and continuities of larger pores appear to be developed by corn root systems, resulting in even greater bypass water flow as the roots decompose. The roots of canola appear to enhance water flow, yet decompose more slowly than corn. Living ryegrass roots actually decrease water flow through wet soils and can be used to catch many of the nutrients during both warm and cold seasons. In conclusion, a ryegrass cover crop can reduce the pollution of groundwater associated with intensive row crop agriculture.

References

Helling, C.S. and T.J. Gish. 1991. Physical and chemical processes affecting preferential flow. In: T.J. Gish and A. Shirmohammadi (ed.) Preferential Flow Proceedings of the National Symposium, 16-17 December, 1991, Chicago Ill. Pub. ASAE, St. Joseph, MI. pp 77-86.

Endnote

[1]Video presentation of "Water Table Management Research in Michigan." 15 minutes.